

Attack–Fault Ambiguity Against a Strategic Attacker: The Stealth–Harm Frontier, Exploitable Commitments, and the Verification Margin

Chiheb Mejrissi

Abstract—Analyses of attack–fault ambiguity at the electronic trading gateway have so far treated the attacker as fixed: its injection is a given alternative to a benign fault, and the operator’s decision rule is designed against it. An attacker who knows the rule can instead choose an injection that imitates the fault. We formulate this as a leader–follower game in a linear-Gaussian order-flow model. The operator commits to a containment rule, possibly with a costly verification step; the attacker, knowing the rule, then chooses a payload whose observation law lies within total-variation distance ε of the fault’s. We ask four questions. How much excess harm, beyond what the imitated fault causes, can an ε -stealthy attacker extract, as a function of ε ? What remains of it as ε tends to zero, under output-level and under state-level injection? Can an attacker exploit a threshold rule committed to a nominal attack model, and what should the operator commit to instead? And what detection margin does a verification step that reads a channel order flow cannot disguise guarantee against such an attacker?

I. INTRODUCTION

An incident at an electronic trading gateway, such as an algorithm that suddenly sends abnormal order flow, can have two causes: a *compromise*, in which an adversary controls the order flow, or a benign *fault*, such as a malfunctioning algorithm. The operator must decide whether to contain the source, at the cost of disrupting a legitimate client if the cause is a fault, or to leave it, at the cost of the harm a compromise goes on to do. Throughout, *the framework* means the setting of Papers 1–3 of this series: an operator observes the order flow over a window, forms the posterior probability that the cause is a compromise, and chooses between containing and leaving, possibly after a costly verification. A *rule* is the operator’s decision procedure in this framework: a map from what it observes (the order flow and, when it verifies, the outcome of the check) to the probability of containing (Definition G5).

Paper 1 [1] models the order flow as a linear-Gaussian system and characterizes when a compromise and a fault cannot be told apart from order flow alone, and what that ambiguity costs the operator. Paper 2 [2] adds verification as a costed action that reads the authorization channel rather than the order flow, which yields a leave/verify/contain rule with two belief thresholds. Paper 3 [3] makes the response sequential, with graded containment. All three treat the attacker as fixed: its

injection is a given alternative to the fault, and the rule is designed against it.

This paper asks whether the framework survives an adversary who knows the rule. We model the interaction as a leader–follower game, as in Stackelberg security games [4]: the operator commits to a rule, and the attacker, knowing it, chooses its injection. Commitment is the natural reading at a trading gateway, where pre-trade controls and kill-switch procedures are documented, tested and audited before any incident; and a rule that is safe only while it is secret offers no security guarantee. The attacker is ε -*stealthy*: the law of the order flow under its injection lies within total-variation distance ε of the law under the fault it imitates.

We address four questions. First, how much excess harm, beyond what the imitated fault causes, can an ε -stealthy attacker with a given payload budget extract, and how does this stealth–harm frontier depend on ε (Section IV)? Second, what remains of the frontier as ε tends to zero, under output-level and under state-level injection? Third, can an attacker exploit a threshold rule committed to a nominal attack model, and what should the operator commit to instead? Fourth, what detection margin does a verification step, which reads a channel order flow cannot disguise, guarantee against a strategic attacker, and where should the operator verify?

Section II positions the paper. Section III states the game, the attacker’s information and the payoffs. Section IV treats the four questions, Section V illustrates them numerically, Section VI discusses simultaneous play and the limits of the model, and Section VII concludes.

II. RELATED WORK

a) *Stealthy attacks on control systems*: Attacks that an output-based monitor cannot detect correspond to the zero dynamics of a linear system and, in the attack-free case, to its unobservable subspace [5]. Paper 1 [1] uses this subspace to characterize when a compromise and a fault remain confounded under state-level injection; this paper takes the mechanism as given and studies its consequence for a strategic attacker. Stealth with a tolerance was introduced as an ε -bound on the Kullback–Leibler divergence between attacked and nominal measurement sequences, with the largest performance degradation given as a function of ε [6]. Here the stealth budget is a total-variation distance, which bounds the detection gap of any

rule directly, and the reference law is that of a benign fault rather than of the attack-free system.

b) Adversarial classification and hypothesis testing:

Adversarial classification casts a classifier and an adversary who modifies its inputs as players of a game [7]. Closest to this paper, game-theoretic Neyman–Pearson detection designs an evasion-aware detector against a strategic evader and summarizes what each side can achieve by an equilibrium receiver operating characteristic [8]. Here the operator minimizes an expected cost with a posterior containment rule instead of a Neyman–Pearson criterion, the benign alternative is a fault, the observations come from a linear dynamical system, and a verification channel is available.

c) Security games: In Stackelberg security games a defender commits to a strategy that the attacker observes before responding, and the relation between Stackelberg and Nash strategies in such games has been studied in detail [4]. This paper adopts the commitment form with the operator as leader and returns to simultaneous play in Section VI.

III. MODEL

This section states the game. The operator, as leader, commits to a rule and an observation window; the attacker, as follower, observes both and chooses an injection within a payload budget; nature then draws the cause, compromise or fault (Definition G1). Observations follow the linear-Gaussian order-flow model of Paper 1 [1], under output-level or state-level injection (Definition G2). The attacker’s payload is the part of its injection that differs from the fault’s (Definition G3), and it is ε -stealthy when the law of the observations under its injection lies within total-variation distance ε of the law under the fault (Definition G4). The operator chooses among rules without verification, including threshold rules committed to a nominal attack model, and rules with a verification step in the sense of Paper 2 [2] (Definition G5). The attacker’s payoff is the expected excess harm it extracts before containment, net of a capture loss (Definition G7); the operator’s loss uses the loss table of Paper 2. The standing assumptions are that the attacker knows the committed rule, the model and the fault instance ((G-A2)), and that verification reads a channel the attacker’s choice of order flow cannot manipulate ((G-A6)).

IV. THE STEALTH–HARM FRONTIER

This section studies the stealth–harm frontier $\mathcal{F}_W(\varepsilon)$, the largest excess harm an ε -stealthy attacker with a given payload budget can extract on the window W (Definition G9). It bounds the detection probability of an ε -stealthy attacker against committed rules with and without verification (Proposition R1) and derives the shape of the frontier as a function of ε (Proposition R2). It then derives the harm ceiling, the limit of the frontier as ε tends to zero (Theorem R3), from two results of Paper 1 [1]. Paper 1’s Theorem 1 shows that under output-level injection a compromise and a fault are indistinguishable exactly

when the difference of their injections lies in the kernel of the observation matrix, and its Theorem 2 shows that under state-level injection they are indistinguishable exactly when that difference lies in the unobservable subspace, which is trivial for generic dynamics (its Corollary 3). The section then asks whether an attacker can exploit a threshold rule committed to a nominal attack model (Proposition R4), and what detection margin a verification step guarantees against every ε -stealthy attacker (Proposition R5).

V. NUMERICAL ILLUSTRATION

This section will illustrate the results of Section IV on a small linear-Gaussian instance: the stealth–harm frontier as a function of ε under output-level and state-level injection, and, for rules with verification, the operator’s optimal commitment computed numerically, to compare the Stackelberg-optimal verification region with the non-strategic one of Paper 2 [2].

VI. DISCUSSION

The results are stated for an operator who commits. Under simultaneous play the operator’s rule is a best response to the attacker’s possibly mixed strategy, and its posterior is computed under the attacker’s actual law rather than a nominal one. The discussion separates the results that bound every pair of strategies, and therefore hold under simultaneous play as well, from those that depend on commitment. It also sets out the limits of the model: an attacker who also compromises the authorization channel, for instance with stolen credentials, violates the separate-channel assumption and is out of scope; under an absolute-harm payoff the ceiling is the fault’s own harm rather than zero; and the attacker is assumed to know the fault instance it imitates.

VII. CONCLUSION

This paper extends the attack–fault ambiguity framework to an attacker who knows the operator’s rule. It asks what stealth costs such an attacker in harm, what harm remains when stealth is exact, whether a rule committed to a nominal attack model can be exploited, and how much a verification step on a separate channel protects the operator.

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